



More on the phases of data analysis and this program

Each step in the data analysis process—ask, prepare, process, analyze, share, and act—plays a crucial role in extracting meaningful insights from data. As you navigate through each phase, from asking the right questions to taking informed actions, you harness the true power of data. In this reading, you'll explore how the data analysis process guides this program.

The ask phase

At the start of any successful data analysis, the data analyst:

- Takes the time to fully understand stakeholder expectations
- Defines the problem to be solved
- Decides which questions to answer in order to solve the problem

Qualifying stakeholder expectations means determining who the stakeholders are, what they want, when they want it, why they want it, and how best to communicate with them. Defining the problem means looking at the current state and identifying the ways in which it's different from the ideal state. With expectations qualified and the problem defined, you can derive questions that will help achieve these goals.

In an upcoming course, you'll learn how to ask effective questions and define the problem by working with stakeholders. You'll also cover strategies that can help you share what you discover in a way that keeps people interested.

The prepare phase

In the prepare phase, the **emphasis** is on identifying and locating data you can use to answer your questions. In an upcoming course, you'll learn more about the different types of data and how to identify which kinds of data are most useful for solving a particular problem. You'll also discover why it's so important that data and results are **objective** and **unbiased**. In other words, any decisions made from an analysis should always be based on facts and be fair and impartial.

The process phase

In this phase, the aim is to refine the data. Data analysts find and eliminate any errors and inaccuracies that can get in the way of results. This usually means:

- Cleaning data
- Transforming data into a more useful format
- Combining two or more datasets to make information more complete
- Removing **outliers** (data points that could skew the information)

After data analysts process data, they check the data they prepared to make sure it's complete and correct. This phase is all about getting the details right. Accordingly, the data analyst will refine strategies for verifying and sharing their data cleaning with stakeholders. In an upcoming course, you'll use spreadsheets and structured query language, or SQL, to clean data.

The analyze phase

With a solid foundation of well-defined questions and clean data, you'll **delve** into the analyze phase. This is when you turn the data you've gathered, prepared, and processed into actionable information. Data analysts use many powerful tools in their work. In one upcoming course you'll continue using two of them: spreadsheets and SQL. In another upcoming course you'll explore using the programming language R to work with and analyze data.

The share phase

This phase is exactly what it sounds like: It's time to share what you've learned with your stakeholders! In this part of the program, you'll learn how data analysts interpret results and share them with others to help stakeholders make effective, data-driven decisions. In the share phase, visualization is a data analyst's best friend. So, an upcoming course will highlight why visualization is essential to getting others to understand what your data is telling you. In another upcoming course, you'll learn how to visualize data with R.

The act phase

The data analysis journey culminates in the act phase, when data insights are put to work. For you, this action involves preparing for your job search and having the chance to complete a case study project. It's a great opportunity for you to bring together everything you've worked on throughout this course. Plus, adding a case study to your portfolio helps you stand out from other candidates!

Learn about the process through the program:

1. Learn more about the **ask** phase of the process in the [Ask Questions to Make Data-Driven Decisions](#) course.
2. Learn more about the **prepare** phase of the process in the [Prepare Data for Exploration](#) course.

3. Learn more about the **process** phase of the process in the [Process Data from Dirty to Clean](#) course.
4. Learn more about the **analyze** phase of the process in the [Analyze Data to Answer Questions](#) and [Data Analysis with R Programming](#) courses.
5. Learn more about the **share** phase of the process in the [Share Data Through the Art of Visualization](#) and [Data Analysis with R Programming](#) courses.
6. Learn more about the **act** phase of the process in the [Google Data Analytics Capstone: Complete a Case Study](#) course.

Note: The course links are for you to preview. Please do not complete the courses at this time. You may mark this activity as complete after you understand how the courses align to the data analysis process.